

The Editor-in-Chief
IEEE Transactions on Information Forensics and Security
IEEE Signal Processing Society

Dear Editor-in-Chief,

We respectfully submit our manuscript, “**NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction**,” for consideration as an original research article in *IEEE Transactions on Information Forensics and Security* (TIFS).

Scope alignment. NARCIS is an end-to-end security protocol that transmits authenticated messages by selecting unchanged natural images from a shared index. Its core contributions are at the intersection of information forensics and security: authenticated encryption of payload and metadata, formal security propositions grounded in AES-GCM’s established properties, selection-bias measurement with three detector classes, and systematic evaluation of an attack-qualified finite-index communication system. These contributions fit squarely within TIFS’s focus on cryptographic applications, multimedia security, and forensics.

Key technical contributions.

- **Feasible operating point (Contribution 1).** A codebook is accepted only when every attack-qualified bucket can satisfy the multiplicity required by every protected coded message. The selected alphabet is the largest satisfying this criterion. Over 1,575 end-to-end message-condition trials on BOSSBase 1.01 and Caltech-101, NARCIS recovered 100% of authenticated plaintexts under 21 channel conditions including eight unseen holdout strengths.
- **End-to-end authenticated protocol (Contribution 2).** AES-GCM (NIST SP 800-38D) encrypts the payload; associated data binds sender and receiver identities. A monotonic sequence counter rejects replay. A keyed Gray permutation limits public symbol-to-label correspondence. Reed–Solomon forward error correction handles residual label errors. Two formal propositions (Confidentiality; Authenticity+Freshness) ground these properties in AES-GCM’s IND-CCA2 and authentication guarantee. Net throughput at a 16-symbol operating point scales from 0.184 to 1.213 bits per cover for 8- to 128-byte payloads.
- **Systematic detectability and ablation (Contribution 3).** Three detector classes (global logistic regression, residual-feature ExtraTrees, selection CNN) are evaluated on five partitions of each dataset. BOSSBase AUCs include chance. Caltech-101 shows a weak but reproducible partition-specific CNN signal (20-repeat mean AUC 0.533, 95% CI [0.517, 0.548]), reported transparently. Removing attack qualification reduces BOSSBase seed-11 complete-message success from 210/210 to 142/210 (67.6%), with crop and large-rotation conditions failing completely, demonstrating that the qualification filter is the dominant robustness mechanism.

Novelty and positioning. Selection-based coverless systems have been evaluated at the descriptor level (average symbol accuracy, nominal capacity) but have not integrated cryptographic payload protection, replay control, or error correction into a single verifiable system. NARCIS introduces the *feasible-codebook* criterion as the primary evaluation unit, closes this protocol gap, and provides matching detectability and ablation evidence. The paper explicitly positions itself relative to related methods (Abdulsattar 2021; Guo–Ping 2026; Ren–Wu 2024) without overstating numerical comparisons across different datasets and protocols.

Author declarations.

- The manuscript is original, has not been published, and is not under consideration elsewhere. It was previously desk-rejected by JISA (Ms. No. JISAS-D-26-03622) for scope reasons; this submission has been substantially revised and is not a transfer.
- All four authors approved the final manuscript and author order. WYSAWIS (cited in related work) shares authors with this submission and is identified explicitly in the paper; it is used as qualitative context, not as an experimental baseline.
- No competing interests and no external funding.
- No human subjects, personal data, or animals are involved. Implementation, fixed seeds, experimental results, and manuscript sources are available at github.com/EkodeckStephane/NARCIS.

We believe NARCIS will interest TIFS readers working on authenticated multimedia communication, selection forensics, and practical steganographic security systems. We welcome the opportunity to provide any additional information the Associate Editor may require.

Yours sincerely,

Stéphane Gaël R. Ekodeck (Corresponding Author)
ORCID: 0000-0002-8094-8832

Co-authors: Vivien Beyala Kamgang (ORCID 0000-0002-3644-1866); Serge Alain Ebélé (ORCID 0000-0001-7036-7068); Julius Marcellin Nkenlifack (ORCID 0000-0003-3322-4687).